

spark_job_config

JOB_BACKEND_IMPL module-attribute

```
JOB_BACKEND_IMPL = autoclass('com.feedzai.pulse.service.datascience.config.JobConfigurator$JobBackendImplementation')
```

JOB_TYPE module-attribute

```
JOB_TYPE = autoclass('com.feedzai.pulse.service.datascience.JobType')
```

KEY_TAB_DIR module-attribute

```
KEY_TAB_DIR = 'KEY_TAB_DIR'
```

logger module-attribute

```
logger = getLogger(__name__)
```

SparkJobConfig

```
SparkJobConfig(pyjnius_java)
```

Bases: [JavaWrapper](#)

Represents a spark job configuration.

subclasses class-attribute instance-attribute

```
subclasses: List[JavaWrapper] = []
```

create classmethod

```
create(sparkJobName, sparkConfig, kerberosConfig)
```

SparkMasterURL dataclass

```
SparkMasterURL(master: str, deploy_mode: str | None)
```

Stores information about the Spark master URL.

This class is useful to parse master URLs from Pulse Configuration, while maintaining compatibility with different Spark versions/distributions.

See more at [Submitting Applications / Master URLs](#)

Attributes:

Name	Type	Description
master	str	Spark master URL. One of 'local', 'spark', 'mesos', 'yarn', or 'k8s'.

Name	Type	Description
<code>deploy_mode</code>	(str, optional)	Deploy mode for the different master URLs. One of 'client' or 'cluster'.

deploy_mode instance-attribute [¶](#)

```
deploy_mode: str | None
```

master instance-attribute [¶](#)

```
master: str
```

from_pulse_config classmethod [¶](#)

```
from_pulse_config(config: Any) -> SparkMasterURL
```

Return the master URL compatible with Spark 2 onwards.

Previous URLs would be called local, yarn-client, yarn-cluster. However, Spark 2 introduced a "special" Spark master which is just YARN and delegates the deployment mode to Hadoop. The new options are::

```
--master yarn --deploy-mode client|cluster
```

This function translates 'yarn-client' and 'yarn-cluster' to their respective Spark 2+ configs.

Parameters:

Name	Type	Description	Default
<code>config</code> ¶	SparkConfig	Spark Config as returned by :func: get_spark_session_with_app_name or :func: get_spark_session_without_app_name .	<i>required</i>

add_kerberos_to_spark_properties [¶](#)

```
add_kerberos_to_spark_properties(pulse: Server, spark_config: dict) -> None
```

Adds kerberos configuration to `spark_config` .

Note

This function mutates the contents of `spark_config` .

get_spark_config_with_app_name [¶](#)

```
get_spark_config_with_app_name(pulse: Server, app_name: str | None, job_profile_name: str | None = None, group_name: str | None = None, clean_spark_properties: bool = True) -> Any
```

Retrieves the effective 'com.feedzai.pulse.service.datascience.config.SparkConfig' from Pulse Server to be used by an external Spark job. These configurations will depend on a given optional job profile and group model.

Parameters:

Name	Type	Description	Default
<code>pulse</code> ¶	<code>Server</code>	The ds_api.Server used to communicate with Pulse.	<i>required</i>
	str		<i>required</i>

Name	Type	Description	Default
<code>app_name</code> ↗		An optional name of a pulse application. This parameter should be used when we want to have access to application specific properties. This allows to use the same properties being used by Spark jobs executed in Pulse.	
<code>job_profile_name</code> ↗	str	The name of the job profile to reuse the Spark properties defined there. If 'None' then it will retrieve the properties associated to default job profile for jobs of type 'EXTERNAL_DS_JOB'.	None
<code>group_name</code> ↗	str	The name of the group model used to retrieve the configurations defined in its properties.	None
<code>clean_spark_properties</code> ↗	bool	True if it should clean the Spark properties specific to Pulse use case, false otherwise.	True

Returns:

Type	Description
SparkConfig	A com.feedzai.pulse.service.datascience.config.SparkConfig with Pulse's configuration.

get_spark_config_without_app_name [↗](#)

```
get_spark_config_without_app_name(pulse: Server) -> Any
```

Retrieves the effective 'com.feedzai.pulse.service.datascience.config.SparkConfig' from configurations defined in the global scope of 'com.feedzai.pulse.service.datascience.config.DataScienceConfig' and com.feedzai.pulse.service.distributed.job.config.DistributedJobServiceConfig .

The DataScienceConfig has priority over DistributedJobServiceConfig.

Parameters:

Name	Type	Description	Default
<code>pulse</code> ↗	Server	The ds_api.Server used to communicate with Pulse.	<i>required</i>

Returns:

Type	Description
unknown	A 'com.feedzai.pulse.service.datascience.config.SparkConfig' from configurations defined in the global scope of data science service configuration.